

Independent Reproof of the Mahar–Willner Doubling Lemmas (Cell Extension and Zero-Truncation Induction, with the Boundary-Slope Transmission Coefficients)

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Abstract

We give a fully self-contained proof of the two structural lemmas used by Mahar–Willner [1] in their solution of the extremal problem for λ_2/λ_1 of $-y'' = \lambda \rho y$ (Dirichlet, $a \leq \rho \leq 1$): Lemma 1 (the *cell-extension* identity $\lambda_{2n}(\rho_n)/\lambda_n(\rho_n) = \lambda_2(\rho_0)/\lambda_1(\rho_0)$ for the $1/n$ -periodic “blow-up” ρ_n of an arbitrary ρ_0) and Lemma 2 (the *zero-truncation* induction establishing $\mu_{2n,n}(a) = \mu(a)$, $\nu_{2n,n}(a) = \nu(a)$). The only external facts used are the classical Sturm oscillation and comparison theorems. In the minimum case we give a variant of the truncation argument which is cleaner than the “similarly” remark in [1]: the side on which one truncates and the direction of the shooting are dictated by the position of z_0 relative to z_{2k} , and the induction hypothesis is needed only in one of the two cases. The proof makes the sup-ratio theorem $\sup_{n,\rho} \lambda_{n+1}/\lambda_n = \nu(R)$ of the accompanying report *fully self-contained*.

1 Setting and notation

Throughout, ρ denotes a piecewise constant density on $[-1/2, 1/2]$ with $a \leq \rho \leq 1$ pointwise ($0 < a < 1$), and $\mathcal{P}_a$ is the class of such densities. Consider

$$-y'' = \lambda \rho(x) y, \quad y(-1/2) = y(1/2) = 0, \quad (1)$$

with eigenvalues $0 < \lambda_1(\rho) < \lambda_2(\rho) < \dots$ (simplicity and discreteness are classical). Write

$$\mu(a) := \inf_{\rho \in \mathcal{P}_a} \frac{\lambda_2}{\lambda_1}, \quad \nu(a) := \sup_{\rho \in \mathcal{P}_a} \frac{\lambda_2}{\lambda_1}, \quad \mu_{2n,n}(a) := \inf_{\rho \in \mathcal{P}_a} \frac{\lambda_{2n}}{\lambda_n}, \quad \nu_{2n,n}(a) := \sup_{\rho \in \mathcal{P}_a} \frac{\lambda_{2n}}{\lambda_n}.$$

The eigenfunction of λ_j is denoted y_j ; it is normalized arbitrarily, and we always take $y'_j(-1/2) \neq 0$ (true up to rescaling).

Preliminary facts. We shall use the following classical results; proofs are sketched only where the formulation is non-standard.

Proposition 1.1 (affine invariance of ratios). *Let $L(x) = cx + d$ with $c \neq 0$ map $[x_0, x_1]$ onto $[-1/2, 1/2]$. If $\lambda_1 < \lambda_2$ are the first two Dirichlet eigenvalues of (1) on $[L(x_0), L(x_1)]$ with density ρ , then the first two Dirichlet eigenvalues of $y'' + \lambda \rho(L(x))y = 0$ on $[x_0, x_1]$ are λ_1/c^2 , λ_2/c^2 . In particular the ratio λ_2/λ_1 is invariant.*

Proposition 1.2 (Sturm oscillation). *The j -th eigenfunction y_j of (1) has exactly $j - 1$ (interior) zeros in $(-1/2, 1/2)$, all simple.*

Proposition 1.3 (Sturm interlacing). *If $j < k$ and $z_1 < \dots < z_{j-1}$ are the zeros of y_j , then y_k has at least one zero in each of the j intervals $(-1/2, z_1), (z_1, z_2), \dots, (z_{j-1}, 1/2)$.*

The backbone of the truncation argument is the following elementary monotonicity of the zeros of an initial-value problem (IVP).

Proposition 1.4 (zero motion). *Fix $c \neq 0$ and let $y(\cdot; \lambda)$ be the solution of*

$$y'' + \lambda \rho(x)y = 0, \quad y(-1/2; \lambda) = 0, \quad y'(-1/2; \lambda) = c.$$

For $j \geq 1$ let $z_j(\lambda)$ be the j -th zero of $y(\cdot; \lambda)$ to the right of $-1/2$, whenever it exists. Then each z_j is continuous and strictly decreasing in λ . The analogous statement holds for the IVP posed at $x = 1/2$: the j -th zero to the left of $1/2$ is strictly increasing in λ .

Proof. Continuity is standard (smooth dependence on the parameter λ for ODEs with simple zeros). For the monotonicity, use the Prufer representation $y = r \sin \theta$, $y' = \sqrt{\lambda \rho} r \cos \theta$, $\theta(-1/2) = 0$. Then θ satisfies $\theta' = \sqrt{\lambda \rho} \cos^2 \theta + \frac{(\sqrt{\lambda \rho})'}{\sqrt{\lambda \rho}}$. (the exact form is immaterial); the comparison theorem for the Prufer angle gives $\theta(x; \lambda_1) < \theta(x; \lambda_2)$ for $\lambda_1 < \lambda_2$ and every fixed x (strictly, since $\sqrt{\lambda_2 \rho} - \sqrt{\lambda_1 \rho} > 0$ a.e.). The zeros of $y(\cdot; \lambda)$ are the solutions of $\theta(x; \lambda) = j\pi$, so $z_j(\lambda_2) < z_j(\lambda_1)$. The statement at $x = 1/2$ follows by the reflection $x \mapsto 1/2 - x$. $\square$

An equivalent formulation, used repeatedly below, is:

Corollary 1.5. *Let $\bar{y}(\cdot; \lambda)$ be an IVP solution as above and let $x_* \in (-1/2, 1/2)$ be a fixed point which is not a zero of $\bar{y}(\cdot; \lambda_0)$. As λ increases (resp. decreases) from λ_0 , the value $\bar{y}(x_*; \lambda)$ changes sign exactly when a zero of $\bar{y}$ crosses x_* ; zeros cross one at a time, and the count of zeros of $\bar{y}$ strictly inside a fixed interval whose boundary is x_* changes by ± 1 at each crossing.*

2 Lemma 1: the cell extension

Lemma 2.1 (cell extension). *For every $\rho_0 \in \mathcal{P}_a$, every integer $n \geq 1$, and the $1/n$ -periodic density*

$$\rho_n(x) := \rho_0\left(n\left(x + \frac{1}{2}\right) - \frac{1}{2}\right), \quad -\frac{1}{2} \leq x \leq -\frac{1}{2} + \frac{1}{n}, \quad \text{extended with period } 1/n,$$

one has $\rho_n \in \mathcal{P}_a$ and

$$\frac{\lambda_{2n}(\rho_n)}{\lambda_n(\rho_n)} = \frac{\lambda_2(\rho_0)}{\lambda_1(\rho_0)}. \quad (2)$$

Proof. Fix ρ_0 and write y_1, y_2 for its first two eigenfunctions, with eigenvalues λ_1, λ_2 . Since $y_1 > 0$ on $(-1/2, 1/2)$, $y_1(\pm 1/2) = 0$, we have $y_1'(-1/2) > 0 > y_1'(1/2)$; define the transmission coefficient

$$s := -\frac{y_1'(1/2)}{y_1'(-1/2)} > 0.$$

Similarly y_2 has exactly one interior zero, so its boundary slopes have opposite signs and $t := -y_2'(1/2)/y_2'(-1/2) > 0$ satisfies $y_2'(1/2) = -t y_2'(-1/2)$.

Set $x_j := -1/2 + j/n$ and, for $x \in [x_j, x_{j+1}]$, $u_j(x) := n(x - x_j) - 1/2 \in [-1/2, 1/2]$. Define

$$\tilde{y}(x) := (-s)^j y_1(u_j(x)), \quad \tilde{f}(x) := (-t)^j y_2(u_j(x)), \quad x \in [x_j, x_{j+1}], \quad j = 0, \dots, n-1.$$

Both are C^1 across every cell boundary x_{j+1} : continuity holds because $y_1(\pm 1/2) = y_2(\pm 1/2) = 0$, and the slopes match because

$$(-s)^j y_1'(1/2) = (-s)^{j+1} y_1'(-1/2), \quad (-t)^j y_2'(1/2) = (-t)^{j+1} y_2'(-1/2).$$

On cell j one has $\tilde{y}'' = -n^2\lambda_1\rho_n\tilde{y}$ and $\tilde{f}'' = -n^2\lambda_2\rho_n\tilde{f}$, so both are eigenfunctions of (1) with $\rho = \rho_n$, of eigenvalues $n^2\lambda_1$ and $n^2\lambda_2$. Their boundary values at $\pm 1/2$ vanish.

Zeros: $\tilde{y}$ vanishes exactly at the cell boundaries, i.e. at $n - 1$ interior points, hence it is the n -th eigenfunction (Proposition 1.2) and $\lambda_n(\rho_n) = n^2\lambda_1(\rho_0)$. The function $\tilde{f}$ vanishes at the $n - 1$ cell boundaries and at one point inside each cell (the image of the interior zero of y_2), i.e. at $2n - 1$ interior points, hence it is the $2n$ -th eigenfunction and $\lambda_{2n}(\rho_n) = n^2\lambda_2(\rho_0)$. Dividing gives (2). $\square$

Corollary 2.2. $\mu_{2n,n}(a) \leq \mu(a)$ and $\nu_{2n,n}(a) \geq \nu(a)$ for all $n \geq 1$.

Proof. For any ρ_0 , (2) says that the value $\lambda_2(\rho_0)/\lambda_1(\rho_0)$ is attained by the ratio λ_{2n}/λ_n at $\rho_n \in \mathcal{P}_a$. Taking the supremum (resp. infimum) over ρ_0 gives $\nu_{2n,n}(a) \geq \nu(a)$ (resp. $\mu_{2n,n}(a) \leq \mu(a)$). $\square$

3 Lemma 2: the zero-truncation induction

Throughout this section, $k \geq 1$, $m := k + 1$, and ρ is an arbitrary element of $\mathcal{P}_a$. We write

$$z_0 := \text{rightmost interior zero of } y_m, \quad z_1 := \text{second-from-right interior zero of } y_{2m},$$

and $z_1 < z_2 < \dots < z_{2k+1}$ for all interior zeros of y_{2m} (so $z_{2k} = z_1$). By Proposition 1.3, y_{2m} has at least one zero in each of the k intervals between consecutive zeros of y_m inside $(-1/2, z_0]$; in particular at least k zeros of y_{2m} lie strictly inside $(-1/2, z_0)$ and at least one lies in $(z_0, 1/2)$.

3.1 The maximum case

Lemma 3.1 (truncation, max). *For every $\rho \in \mathcal{P}_a$ and every $k \geq 1$,*

$$\frac{\lambda_{2(k+1)}(\rho)}{\lambda_{k+1}(\rho)} \leq \max\{\nu_{2k,k}(a), \nu(a)\}.$$

Proof. We fix ρ and suppress it from the notation. *Case A:* $z_0 \leq z_1$ (i.e. $z_0 \leq z_{2k}$). Let $\bar{y}(\cdot; \lambda)$ be the IVP solution $y'' + \lambda\rho y = 0$, $y(-1/2; \lambda) = 0$, $y'(-1/2; \lambda) = 1$, and let j be the number of zeros of y_{2m} strictly inside $(-1/2, z_0)$. Since $z_{2k} \geq z_0$, we have $j \leq 2k - 1$. By Proposition 1.4 and Corollary 1.5, there is a smallest $\bar{\lambda} \geq \lambda_{2m}$ with $\bar{y}(z_0; \bar{\lambda}) = 0$; at this value the zero of $\bar{y}$ lying just to the right of z_0 at $\lambda = \lambda_{2m}$ has moved to z_0 , so $\bar{y}(\cdot; \bar{\lambda})$ has exactly $j \leq 2k - 1$ zeros strictly inside $(-1/2, z_0)$.

Consider the truncated problem $y'' + \lambda\rho y = 0$ on $(-1/2, z_0)$ with $y(-1/2) = y(z_0) = 0$. Its k -th eigenfunction is y_m restricted ($k - 1$ zeros inside), with eigenvalue λ_{k+1} ; its $(j + 1)$ -st eigenfunction is $\bar{y}(\cdot; \bar{\lambda})$ (j zeros inside), with eigenvalue $\bar{\lambda}$. Since $j \geq k$ (Proposition 1.3), $k < j + 1 \leq 2k$, and $\lambda_{j+1} \leq \lambda_{2k}$ pointwise, the affine normalization (Proposition 1.1) gives

$$\frac{\bar{\lambda}}{\lambda_{k+1}} \leq \sup_{\rho' \in \mathcal{P}_a} \frac{\lambda_{j+1}(\rho')}{\lambda_k(\rho')} \leq \sup_{\rho' \in \mathcal{P}_a} \frac{\lambda_{2k}(\rho')}{\lambda_k(\rho')} = \nu_{2k,k}(a).$$

Since $\lambda_{2m} \leq \bar{\lambda}$, we obtain $\lambda_{2m}/\lambda_{k+1} \leq \nu_{2k,k}(a)$.

Case B: $z_1 < z_0$ (i.e. $z_{2k} < z_0$). Now y_{2m} has exactly one zero, z_{2k+1} , strictly inside $(z_0, 1/2)$. Let $\bar{y}(\cdot; \lambda)$ be the IVP solution posed at $x = 1/2$: $y(1/2; \lambda) = 0$, $y'(1/2; \lambda) = 1$. As λ increases, its zeros move toward $1/2$; there is a smallest $\bar{\lambda} \geq \lambda_{2m}$ with $\bar{y}(z_0; \bar{\lambda}) = 0$, and at this value $\bar{y}$ has exactly one zero strictly inside $(z_0, 1/2)$ (namely the image of z_{2k+1}). Hence $\bar{\lambda}$ is the second eigenvalue of the truncated problem on $(z_0, 1/2)$ whose first eigenfunction is y_m restricted (no zeros inside), of eigenvalue λ_{k+1} . Normalizing affinely,

$$\frac{\bar{\lambda}}{\lambda_{k+1}} \leq \nu(a), \quad \text{so} \quad \frac{\lambda_{2m}}{\lambda_{k+1}} \leq \nu(a).$$

In both cases $\lambda_{2m}/\lambda_{k+1} \leq \max\{\nu_{2k,k}(a), \nu(a)\}$, as claimed. $\square$

Theorem 3.2 (MW Lemma 2, max). $\nu_{2n,n}(a) = \nu(a)$ for all $n \geq 1$.

Proof. Induction on k with $n = k+1$. The case $k = 1$ is $\nu_{2,1}(a) = \nu(a)$ by definition. If $\nu_{2k,k}(a) \leq \nu(a)$ (indeed equality holds by Corollary 2.2 and the induction hypothesis), then Lemma 3.1 applied to an arbitrary ρ gives $\lambda_{2(k+1)}/\lambda_{k+1} \leq \nu(a)$; taking the supremum over ρ yields $\nu_{2(k+1),(k+1)}(a) \leq \nu(a)$. The reverse inequality is Corollary 2.2. $\square$

3.2 The minimum case

For the minimum we use the following variant, which is the precise form of the “minimization following similarly” in [1]. Let c_0 be the number of zeros of y_{2m} strictly inside $(z_0, 1/2)$. By interlacing $c_0 \geq 1$.

Lemma 3.3 (truncation, min). *For every $\rho \in \mathcal{P}_a$ and every $k \geq 1$,*

$$\frac{\lambda_{2(k+1)}(\rho)}{\lambda_{k+1}(\rho)} \geq \min\{\mu_{2k,k}(a), \mu(a)\}.$$

Proof. Fix ρ . *Case A:* $c_0 \geq 2$ (i.e. $z_0 < z_{2k}$). Consider the truncated problem on $(z_0, 1/2)$; its first eigenfunction is y_m restricted, of eigenvalue λ_{k+1} . Let $\bar{y}(\cdot; \lambda)$ be the IVP solution posed at $1/2$ with initial slope 1. As λ decreases from λ_{2m} , the zeros of $\bar{y}$ move to the left (Proposition 1.4) and exit the interval $(z_0, 1/2)$ through z_0 one at a time (Corollary 1.5); the count strictly inside $(z_0, 1/2)$ starts at $c_0 \geq 2$ and decreases by 1 at each crossing. Let $\bar{\lambda}$ be the value at which the count drops from 2 to 1, i.e. the largest $\lambda \leq \lambda_{2m}$ with $\bar{y}(z_0; \lambda) = 0$ after exactly $c_0 - 2$ crossings. Then $\bar{\lambda} \leq \lambda_{2m}$, $\bar{y}(\cdot; \bar{\lambda})$ has exactly one zero strictly inside $(z_0, 1/2)$, and it is an eigenfunction of the truncated problem; hence $\bar{\lambda}$ is its *second* eigenvalue. Normalizing affinely and using the definition of μ ,

$$\frac{\bar{\lambda}}{\lambda_{k+1}} \geq \mu(a), \quad \text{so} \quad \frac{\lambda_{2m}}{\lambda_{k+1}} \geq \mu(a).$$

Case B: $c_0 = 1$ (i.e. $z_{2k} < z_0$). Then all $2k$ zeros $z_1, \dots, z_{2k}$ of y_{2m} lie strictly inside $(-1/2, z_0)$. Let $\bar{y}(\cdot; \lambda)$ be the IVP solution posed at $-1/2$ with initial slope 1. As λ decreases, zeros move to the right; let $\bar{\lambda}$ be the largest $\lambda \leq \lambda_{2m}$ such that $\bar{y}(z_0; \lambda) = 0$ (the first crossing). Then $\bar{y}(\cdot; \bar{\lambda})$ has exactly $2k - 1$ zeros strictly inside $(-1/2, z_0)$, so it is the $2k$ -th eigenfunction of the truncated problem on $(-1/2, z_0)$, while y_m restricted is its k -th eigenfunction with eigenvalue λ_{k+1} . Hence, after affine normalization,

$$\frac{\bar{\lambda}}{\lambda_{k+1}} \geq \mu_{2k,k}(a), \quad \text{so} \quad \frac{\lambda_{2m}}{\lambda_{k+1}} \geq \mu_{2k,k}(a).$$

In both cases the desired lower bound holds. $\square$

Theorem 3.4 (MW Lemma 2, min). $\mu_{2n,n}(a) = \mu(a)$ for all $n \geq 1$.

Proof. Induction on k with $n = k + 1$. Base case $k = 1$: $\mu_{2,1}(a) = \mu(a)$ by definition. If $\mu_{2k,k}(a) \geq \mu(a)$, then Lemma 3.3 applied to an arbitrary ρ gives $\lambda_{2(k+1)}/\lambda_{k+1} \geq \mu(a)$; taking the infimum yields $\mu_{2(k+1),(k+1)}(a) \geq \mu(a)$. The reverse inequality is Corollary 2.2. $\square$

4 Consequences

Theorem 4.1 (MW Theorem 3). *If ρ_0 extremizes λ_2/λ_1 in $\mathcal{P}_a$, then the cell extension ρ_n of Lemma 2.1 extremizes λ_{2n}/λ_n . Consequently*

$$\mu_{2n,n}(a) = \mu(a), \quad \nu_{2n,n}(a) = \nu(a) \quad (n \geq 1).$$

Proof. If ρ_0 maximizes, then $\nu(a) = \lambda_2(\rho_0)/\lambda_1(\rho_0) = \lambda_{2n}(\rho_n)/\lambda_n(\rho_n) \leq \nu_{2n,n}(a)$ by definition, and Theorem 3.2 gives the reverse inequality; hence ρ_n maximizes λ_{2n}/λ_n . The minimum case is analogous with Theorem 3.4. $\square$

Corollary 4.2 (self-contained sup-ratio theorem). *Let $R > 1$ and $0 < a \leq \rho \leq A = aR$. For the problem $-y'' = \lambda \rho y$ (Dirichlet on $[-1/2, 1/2]$),*

$$\sup_{n \geq 1} \sup_{\rho} \frac{\lambda_{n+1}}{\lambda_n} = \nu(R) = \left(\frac{\arccos(-\sqrt{R}/(\sqrt{R}+1))}{\arccos(\sqrt{R}/(\sqrt{R}+1))} \right)^2,$$

attained at $n = 1$ and the symmetric two-jump configuration $[a, A, a]$.

Proof. By scaling $\rho \mapsto \rho/a$ we may assume $a = 1$, $A = R$. The upper bound: $\lambda_{n+1} \leq \lambda_{2n}$ and Theorem 3.2 give $\lambda_{n+1}/\lambda_n \leq \nu_{2n,n}(R) = \nu(R)$. The lower bound and the closed formula for $\nu(R)$, together with the explicit extremizer $[1, R, 1]$ with block widths st, t, st , $s = \sqrt{R}$, $t = 1/(2s+1)$, are established in the accompanying report `SL_ratio_proof.pdf` (balanced-phase computation). This proves the claim without appealing to [1, Lemmas 1–2]. $\square$

5 Numerical audit

All computations use exact piecewise-constant transfer-matrix propagation (double precision) with bisection for eigenvalues and zero locations.

1. *Lemma 2.1.* For 8 random densities ρ_0 with 2–4 jumps and $1 \leq \rho_0 \leq 4$, the identity (2) was verified for $n = 2, 3$ to relative error $\leq 4.5 \times 10^{-16}$. Additionally the identities $\lambda_n(\rho_n) = n^2 \lambda_1(\rho_0)$, $\lambda_{2n}(\rho_n) = n^2 \lambda_2(\rho_0)$ were checked to 6 digits, confirming the C^1 pasting structure. (Script `op05_mw_lemma12_audit.py`.)
2. *Lemma 3.1.* For 24 random configurations (10 case A, 14 case B, $k = 1, 2, 3$), the shooting value $\bar{\lambda}$ was computed by sign-change bisection and verified to satisfy: case A — $\bar{\lambda}$ is the $(j+1)$ -st eigenvalue of the truncated problem with $j \leq 2k-1$ zeros, $j \geq k$, and $\bar{\lambda} \geq \lambda_{2m}$; case B — $\bar{\lambda}$ is the second eigenvalue of the $(z_0, 1/2)$ -problem, with exactly one interior zero, and $\bar{\lambda} \geq \lambda_{2m}$. All checks passed to relative error $< 10^{-4}$ (limited by the zero-location precision of z_0). (Script `op05_mw_lemma2_audit.py`.)
3. *Lemma 3.3.* For 20 random configurations (14 case A, 6 case B, $k = 1, 2, 3$), case A: the shooting value equals the second eigenvalue of $(z_0, 1/2)$ and $\bar{\lambda} \leq \lambda_{2m}$ (with ratios $e_2/\lambda_m \in [3.65, 4.18]$); case B: $\bar{\lambda}$ equals the $(2k)$ -th eigenvalue of $(-1/2, z_0)$ and $\bar{\lambda} \leq \lambda_{2m}$. All checks passed. (Scripts `op05_mw_mincase_audit2.py`.)
4. *Zero motion (Proposition 1.4).* For 5 random densities, the ordered zero positions of the IVP solution at λ and 1.05λ are strictly ordered as predicted. (Script `op05_mw_extra_checks.py`.)

6 Audit notes

- The case split in Lemma 3.1 is exhaustive: $z_0 \leq z_1$ or $z_1 < z_0$. The degenerate case $z_0 = z_1$ is covered by case A with $j \leq 2k-1$ (then $z_{2k} = z_0$ and $\bar{\lambda} = \lambda_{2m}$ is admissible).
- The identity $j \geq k$ in case A of Lemma 3.1 follows from interlacing: each of the k intervals between $-1/2$ and the successive zeros of y_m contains a zero of y_{2m} strictly inside $(-1/2, z_0)$.

- In case A of Lemma 3.3 the “count drops from 2 to 1” uses the monotonicity of the zero count in λ ; the value $\bar{\lambda}$ is obtained after exactly $c_0 - 2$ crossings, all of which occur at $\lambda < \lambda_{2m}$, so $\bar{\lambda} \leq \lambda_{2m}$.
- We do not need the existence of extremizers for the inequalities $\nu_{2n,n} \leq \nu$ and $\mu_{2n,n} \geq \mu$: Lemma 3.1 (resp. Lemma 3.3) is proved for an *arbitrary* $\rho \in \mathcal{P}_a$, and only the final supremum (resp. infimum) is taken. This removes the compactness question entirely.
- The transmission coefficients s, t are only used in Lemma 2.1; the equality $s = t$ is *not* required for the doubling identities (it is used in [1] only to prove the symmetry of the extremizer, Theorem 2 there).
- All external facts (oscillation, interlacing, Prufer comparison) are classical; the remaining steps are elementary.

References

- [1] T. J. Mahar, B. E. Willner, *An extremal eigenvalue problem*, Comm. Pure Appl. Math. 29 (1976), no. 5, 517–529. <https://doi.org/10.1002/cpa.3160290505>
- [2] J. B. Keller, *The minimum ratio of two eigenvalues*, SIAM J. Appl. Math. 31 (1976), no. 3, 485–491. <https://doi.org/10.1137/0131042>